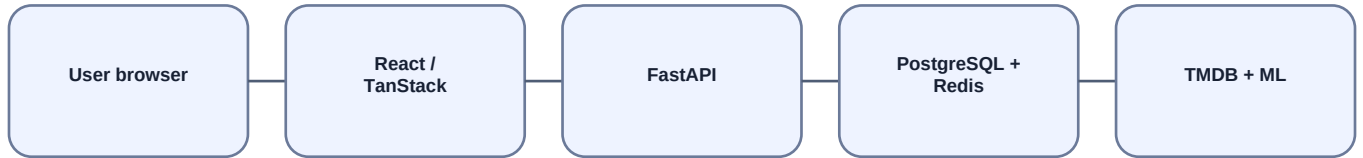


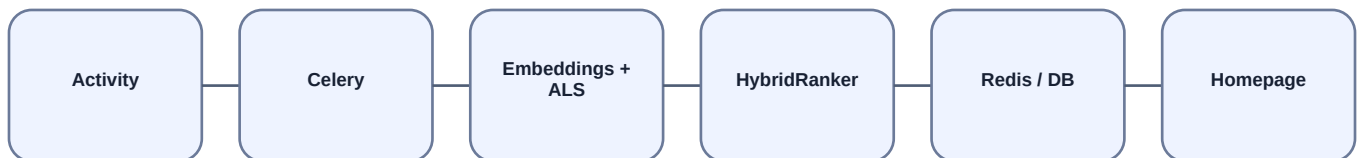
WatchMan Total Workflow & Tech Stack

End-to-end architecture, deployment model and technology inventory

End-to-end flow



Recommendation flow



Production target



Technology stack

Layer	Technology	Role
Frontend	React 19.2	UI.
Framework	TanStack Start + Nitro	SSR/application runtime.
Language	TypeScript 5.8.3	Typed frontend.
Routing	TanStack Router 1.170	File-based routing.
Server state	TanStack React Query 5.101	Fetching/cache.
Client state	Zustand 5.0.14	Client state.
HTTP	Axios 1.18.1	API client.
UI	Radix UI + shadcn wrappers	UI primitives.
CSS	Tailwind CSS 4.2.1	Styling.
Backend	FastAPI	API/services.
Persistence	SQLAlchemy + Alembic	ORM/migrations.
Database	PostgreSQL	Primary store.
Vector	pgvector	Semantic search.

Layer	Technology	Role
Cache/broker	Redis / Upstash Redis	Cache + Celery messaging.
Async	Celery + Celery Beat	Background/scheduled work.
Embeddings	Hugging Face all-MiniLM-L6-v2	384-dim embeddings.
External data	TMDB	Movie/TV metadata.

Secrets and production connection details come from environment configuration. ML-heavy processing runs asynchronously.